Concept note: PEM-based edge inclusion with a standardised VAR core

A redesign of the E8 graph for A3T-GCN | LargeST-GBA 2021, 5-minute flow, 2,056 sensors | 9 November 2026
Status: proposal. Components marked *measured* were tested in a validation-only proxy; the combined design has not been trained yet.

1. Why a redesign

In the first E8 design, a VAR supplied directed, horizon-specific effects, a probabilistic edge model (PEM) supplied an edge probability P for each pair, and the graph scored every edge as $P \odot |\Psi_h|$. PEM therefore *reweighted* VAR edges. Two independent measurements say this did not help forecasting.

1.1 A3T-GCN at the 5-minute horizon (test RMSE, vehicles per 5 min)

Graph	Seed 42	Seed 7	Mean
Road graph	29.76	30.13	29.94
VAR-weighted road graph	28.10	26.71	27.41
E8V: VAR impulse responses, no PEM	26.39	25.87	26.13
E8, uniform prior	25.88	26.02	25.95
E8, shuffled prior (control)	25.86	26.63	26.25
E8, traffic-correlation prior (proposed)	27.30	26.21	26.76
No graph (identity)	27.58	–	–

Every E8 variant beats the road graph on both seeds, by 2.5 to 4.3 RMSE, which is outside the seed-to-seed variation. E8 does **not** beat E8V: the means differ by less than the seed gap of a single variant (0.14 to 1.09, and 1.39 for the VAR-weighted road graph). The criterion fixed in advance, that PEM counts as useful only if E8 beats E8V, is not met.

1.2 Why reweighting cannot help much

- **PEM prunes almost nothing.** Only 0.4% of candidate pairs have $P < 0.01$; 64% have $P \geq 0.5$ and 39% exceed 0.99. The top-25 cap does the pruning.
- **The prior hardly moves P.** With tens of thousands of rows per unit the data outweigh the prior: the three priors give graphs that share 96–98% of their edges at h = 1.
- **P and the VAR edges measure different things.** P is dependence between the *residuals* of two sensors at the same time step; a graph edge is a *lagged* influence. Multiplying them can only rescale VAR edges.
- **The VAR cannot see beyond direct neighbours.** It is restricted to road neighbours, so it cannot represent pairs that share a neighbour but are not adjacent. PEM scores 86,862 such pairs, 63% of its pool.

2. Concept

The redesign keeps the VAR as the directed backbone, fixes how its effects are ranked, and uses PEM for what the VAR cannot do: **include** edges it misses. Figure 1 shows the pipeline.

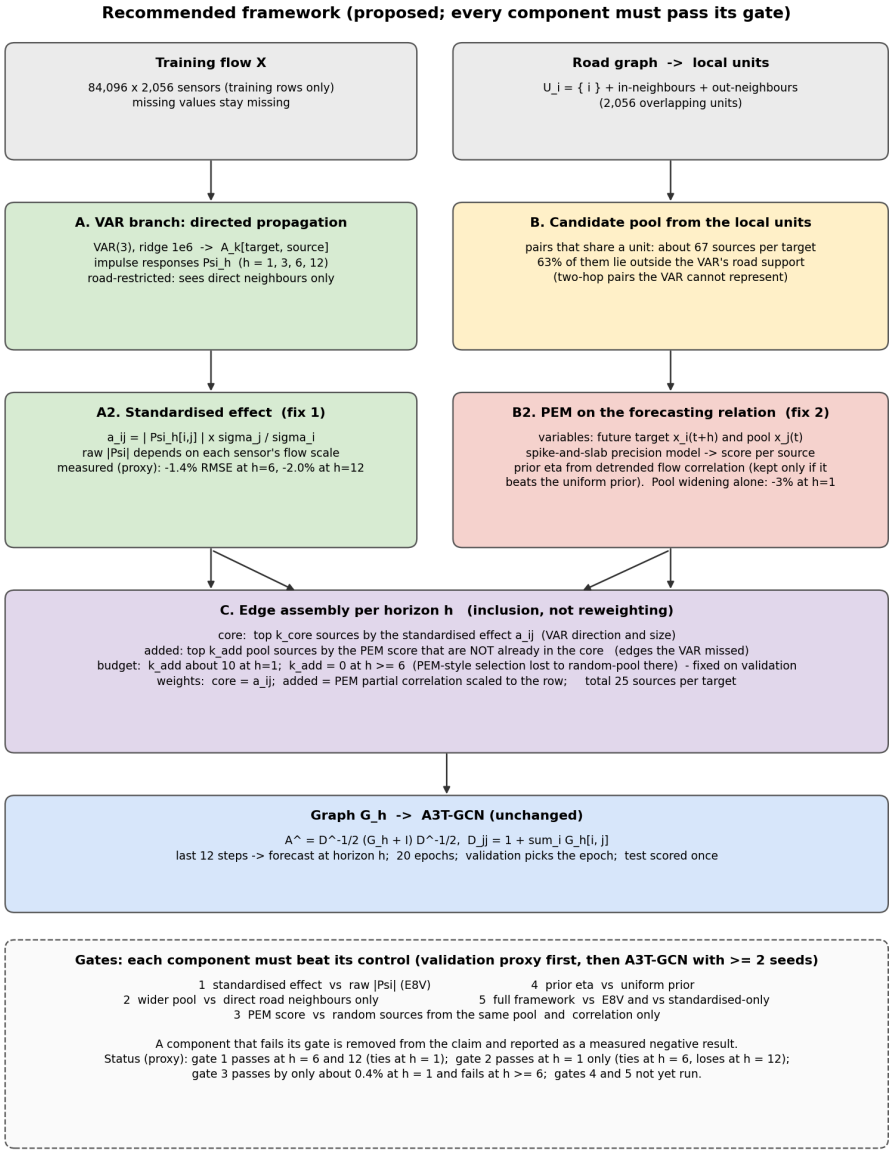


Figure 1. Recommended framework. Every component must pass a gate before it is claimed.

2.1 Branch A: standardised VAR effect (fix 1)

Impulse responses Ψ_h come from the fitted VAR(3), with $\Psi_0 = I$ and $\Psi_s = \sum_{k=1}^{\min(s,3)} A_k \Psi_{s-k}$. Their raw size depends on the flow scale of the two sensors: a high-flow source with a small coefficient can matter more than a low-flow source with a larger one. The edge score is therefore expressed in standardised units,

$$a_h[i,j] \;=\; \left|\Psi_h[i,j]\right| \frac{\sigma_j}{\sigma_i}, \qquad a_h[i,i] = 0, \qquad (1)$$

where σ_j is the standard deviation of the observed *training* flow of sensor j. *Measured (proxy):* this lowers the validation error by 1.4% at h = 6 and 2.0% at h = 12 compared with raw $\left|\Psi_h\right|$, and ties at h = 1. It does not involve PEM.

2.2 Branch B: candidate pool from the local road units

The causal unit of sensor i is $U_i = \{i\} \cup \mathcal{N}_{\text{in}}(i) \cup \mathcal{N}_{\text{out}}(i)$. The pool of candidate sources is every sensor that shares some unit with i:

$$\mathcal{C}_i \;=\; \Big(\bigcup_{U \ni i} U\Big) \setminus \{i\}, \qquad |\mathcal{C}_i| \approx 67. \qquad (2)$$

$\mathcal{C}_i$ contains the direct neighbours the VAR already covers and the two-hop sensors it cannot. *Measured (proxy):* at h = 1, picking 25 sources at random from the pool lowers the validation RMSE by about 3% compared with the VAR core alone.

2.3 PEM on the forecasting relation (fix 2)

The first design applied PEM to residuals, which says nothing about whether a source helps predict a target. Here PEM is applied to the *forecasting relation itself*. For target i and horizon h, form the vector

$$v_i^{(h)}(t) \;=\; \big(\, x_i(t+h),\; x_i(t),\; (x_j(t))_{j \in \mathcal{C}_i} \big), \qquad (3)$$

estimate its precision matrix Ω_i on training rows with the same spike-and-slab objective as before (prior inclusion η_{ij} from the detrended-flow correlation), and score each pool source by its conditional dependence on the future target given all other sources:

$$s_h[i,j] \;=\; \frac{\big|\,\Omega_i[y,j]\,\big|}{\sqrt{\Omega_i[y,y]\;\Omega_i[j,j]}}, \qquad y \equiv x_i(t+h). \qquad (4)$$

Direction comes from time order (future target, current sources); "conditional on all others" separates necessary sources from redundant ones. The spike-and-slab prior turns $\Omega_i[y,j]$ into an inclusion probability, with Laplace densities $f_\nu(x) = \frac{1}{2\nu} e^{-|x|/\nu}$ (spike scale $\nu_0 = 0.01$, slab scale $\nu_1 = 0.2$):

$$P_h^{\text{pred}}[i,j] \;=\; \frac{\eta_{ij}\, f_{\nu_1}\big(\Omega_i[y,j]\big)}{\eta_{ij}\, f_{\nu_1}\big(\Omega_i[y,j]\big) + (1-\eta_{ij})\, f_{\nu_0}\big(\Omega_i[y,j]\big)}. \qquad (5)$$

Caution: so far only a fast stand-in was tested, the same quantity from a shrunk inverse correlation matrix, not the full spike-and-slab estimate.

2.4 Edge assembly: inclusion, not reweighting

With a budget of $K = 25$ sources per target split into $k_{\text{core}} + k_{\text{add}} = K$, the graph keeps

$$G_h[i,j] \;=\; \begin{cases} a_h[i,j] & j \in \operatorname{top}_{k_{\text{core}}}\big(a_h[i,\cdot]\big) \\ \kappa_i\, \rho_h[i,j] & j \in \operatorname{top}_{k_{\text{add}}}\big(s_h[i,\cdot]\big) \text{ over } \mathcal{C}_i \setminus \text{core}_i \\ 0 & \text{otherwise,} \end{cases} \qquad (6)$$

where $\rho_h[i,j]$ is the partial correlation behind $s_h[i,j]$ and κ_i rescales the added weights to the median of the row's core weights, so the two kinds of edge are comparable. PEM-selected edges are **additions** to the VAR backbone. The graph then enters the unchanged A3T-GCN through $\hat{A} = D^{-1/2}(G_h + I)D^{-1/2}$. The budget k_{add} depends on the horizon: about 10 at h = 1 and 0 at h ≥ 6, set on validation data only and frozen before any test run.

3. Evidence so far (validation proxy)

A cheap linear stand-in for A3T-GCN: for each target, a ridge regression predicts its future flow from the last three lags of itself and of the sources the graph gives it. It is fitted on training rows and scored on validation rows only, on 500 random targets; λ is chosen on a training holdout. It ranks graphs; it does not replace the A3T-GCN runs.

Validation RMSE	h = 1	h = 6	h = 12
Self only (no graph)	27.19	43.11	59.45
E8V: raw |Ψ| top-25	26.62	39.98	51.11
E8 flow prior: P × raw |Ψ|	26.64	40.00	51.34
Standardised |Ψ| top-25	26.61	**39.43**	**50.09**
P-ranked inclusion of pool pairs	26.16	39.89	51.42
Random 25 sources from the pool	25.78	**39.43**	50.94
Correlation only, from the pool (no PEM)	25.82	40.16	52.79
Predictive precision (stand-in for 2.3)	**25.69**	41.41	54.92

Reading. (i) Standardising the effect is a clear, PEM-free gain at long horizons. (ii) Widening the pool is the large gain at h = 1 (about 3%), but random pool sources get most of it. (iii) The predictive-precision stand-in is the best graph at h = 1, by only about 0.4% over random pool sources, and the *worst* at 6 and 12, which is why the added budget is 0 there. (iv) P used as a ranker or multiplier does not beat the controls.

4. Gates: how each component earns its place

Gate	Component	Must beat	Status
1	Standardised effect	Raw |Ψ| (E8V)	Passes at h = 6, 12; ties at h = 1 (proxy)
2	Wider pool	Direct neighbours only	Passes at h = 1; ties at h = 6; loses at h = 12 (proxy)
3	PEM score for inclusion	Random pool sources and correlation only	Passes by about 0.4% at h = 1; fails at h ≥ 6 (stand-in)
4	Correlation prior η	Same PEM with a uniform prior	Not yet run
5	Full framework	E8V and standardised-only, ≥ 2 seeds, A3T-GCN	Not yet run

A component that fails its gate is removed from the claim and reported as a measured negative result. A gate at the proxy stage is necessary, not sufficient: only gate 5 involves the real forecaster.

5. Plan and what can be claimed

1. **Hybrid in the proxy** (about 20 minutes, local): core of standardised-effect sources plus pool-selected additions, with the budgets above; sweep k_{add} on validation data only.
2. **If it passes, A3T-GCN at h = 1** with seeds 42 and 7 (about 11 GPU hours) against E8V and the standardised-only graph; the random-pool graph as a control. Longer horizons follow only if validation supports them.
3. **If the PEM score does not beat the random-pool control by more than the seed spread**, the contribution is stated as the standardised effect plus the local-unit candidate pool, with PEM reported as a controlled negative result.

What cannot be promised. The best A3T-GCN run so far scored 25.86 RMSE; the proxy suggests the pool-widened graph might reduce the 5-minute error by 2–3%, which would give roughly 25.3 ± 0.5. Reliably going below 25 would probably also require a more stable training schedule, which affects every model and is a separate decision. Any redesign changes the E8 construction, so it needs approval as an amendment and is reported as post-hoc.

6. Limitations

- The proxy is linear, uses 500 targets and validation data only; transfer to A3T-GCN is unproven.
- Seed-to-seed variation of A3T-GCN is 0.14 to 1.4 RMSE, larger than most proxy margins.
- Only a stand-in for the predictive PEM was evaluated; the spike-and-slab estimate on the forecasting relation (one pass per horizon, about 6 hours of CPU each) has not been run.
- The E8V seed-42 run used TensorFlow 2.21.0 and the other E8 runs 2.20.0.
- Novelty has not been verified beyond a limited web search: time-varying graphs, hour-conditioned graphs and Granger-causality graphs exist in the traffic literature; no paper using VAR impulse responses as adjacency, or PEM-based edge inclusion, was found, but this must be checked before it is claimed.